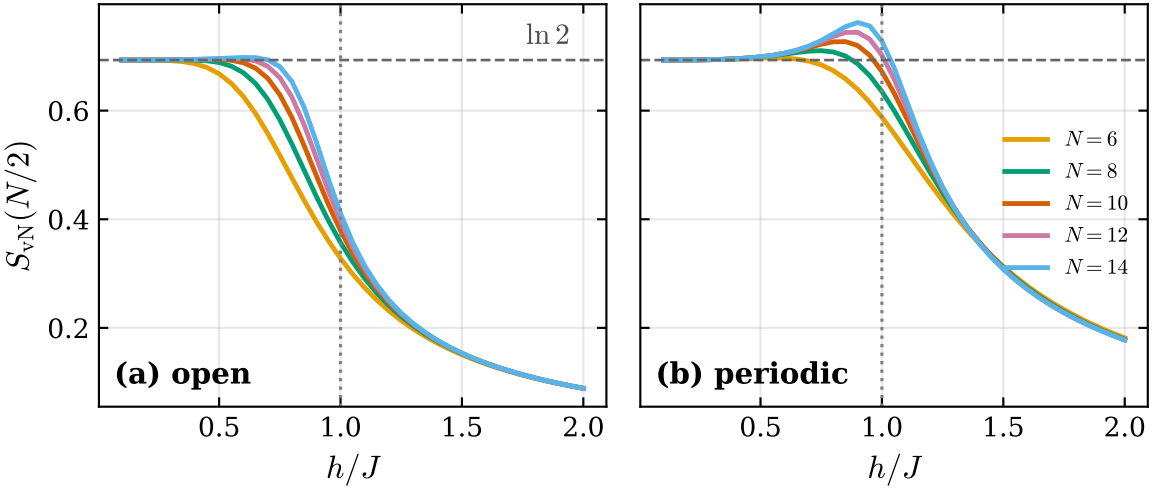
ln 2
0.6
0.4
0.2
$S_{\mathrm{vN}}(N/2)$
(a) open
(b) periodic
0.5
1.0
1.5
2.0
h/J
$N = 6$
$N = 8$
$N = 10$
$N = 12$
$N = 14$